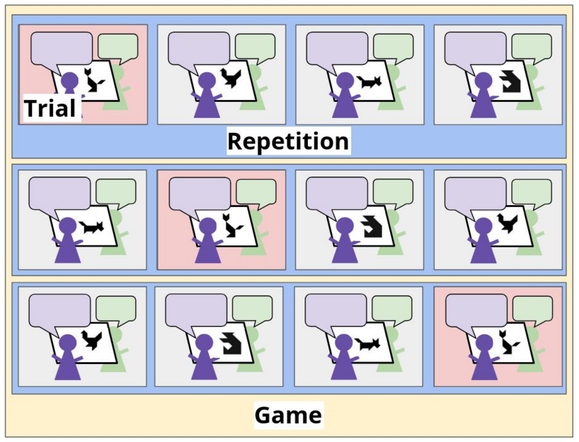
Refbank: an open repository of iterated reference games

Veronica Boyce (vboyce@mit.edu), Alvin W. M. Tan, Jess Mankewitz, Fiona Feng,
Benji Fernandez, Ryder Fried, Ben Prystawski, Michael C. Frank, Robert D. Hawkins

Iterated reference games



Across trials, targets are repeated so that the conversational history shapes the construction and understanding of later utterances.

Why use Refbank?

Convention formation research: Refbank can be used to study moderators and generalizations across datasets, identify gaps to target in future experiments, and build and test theories.

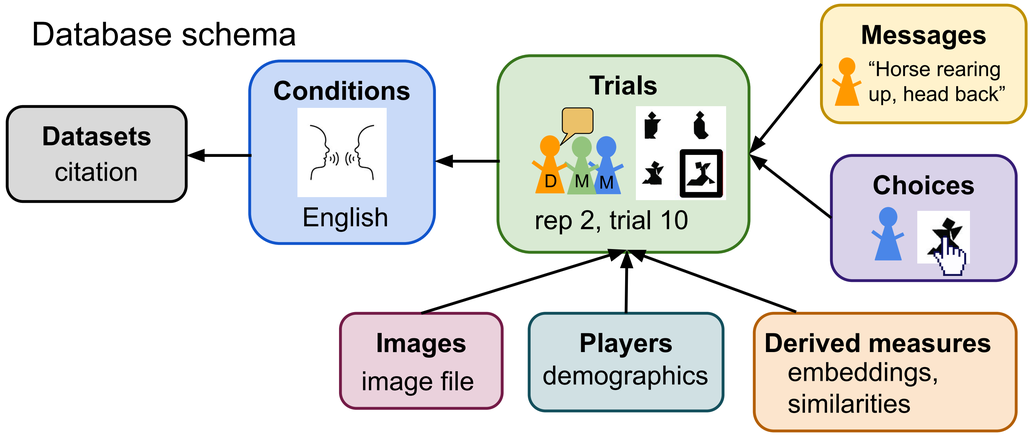
Broader research: Refbank is a corpus of task-specific dialogues that could be relevant for pragmatics, metaphor, syntactic & semantic alignment, and NLP training and benchmarking.

Pedagogy: The corpus is easy to access and could be ideal for class projects or honors theses.

Access Refbank

We present Refbank, an open repository of iterated reference game data.
Interactive visualizations at **`refbank.stanford.edu`**
Access in R using `refbankr` or in Python via the redivis API.

Database schema



Included datasets

Refbank currently has language data from 13 datasets. Unless noted, each game had two adult players.

Paper	#Games	#Trials	#Words	Modality	Lang.	Notes
Branigan et al. (2016)	24	764	30 430	in-person	English	8-10yo children, partner swaps
Yoon & Brown-Schmidt (2019)	788	41 766	348 849	in-person	English	partner swaps, groups of 4-7
Leung et al. (2024)	82	3 259	37 360	in-person	English	parent &4-8yo child
Wang et al. (2025)	20	788	26 105	in-person	English	
Boyce et al. (2025)	97	1 438	16 887	in-person	English	4-5yo children
Bögels et al (2026)	47	4 512	176 483	in-person	Dutch	
Hawkins et al (2019)	168	3 994	19 740	online	English	
Hawkins et al. (2020)	100	7 120	62 590	online	English	
Hawkins et al (2021)	43	3 699	25 777	online	English	groups of 4 in changing pairs
Hawkins et al (2023)	26	2 278	11 756	online	English	groups of 4 in changing pairs
Eliav et al (2023)	262	8 183	50 468	online	English	
Boyce et al. (2024)	306	21 269	374 105	online	English	groups of 2-6
Mankewitz & Hawkins (2025)	447	28 244	198 713	online	English	
Total	2 415	127 401	1 380 037			

Your dataset here!

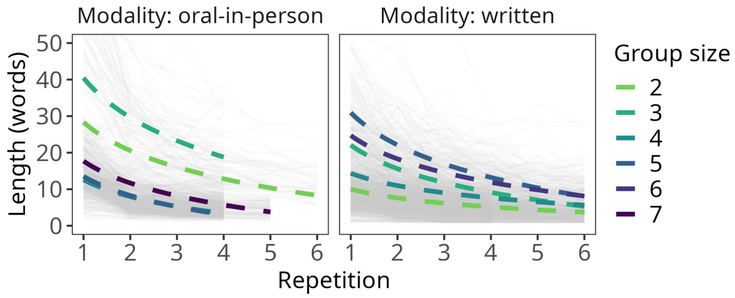
Coming soon!

Paper	#Games	#Trials	#Words	Modality	Lang.	Notes
Bangerter & Smolenski (2000)	18	1 728	20 805	in-person	Swiss German	
Bangerter et al. (2020)	76	3 856	161 253	in-person	French	some have partner swaps
Dahan (2023)	40	1 920	98 006	in-person	English	
Hawkins et al. (unpub)	21	2 268	40 806	in-person	English	

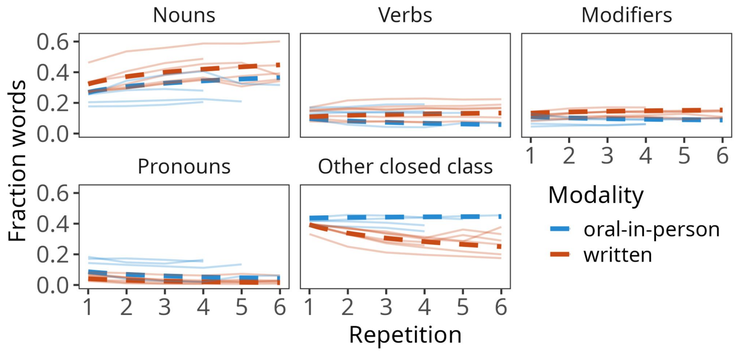
Have you run iterated reference games? Please let us know if you have a dataset you would like to contribute!

Moderators and Generalizations

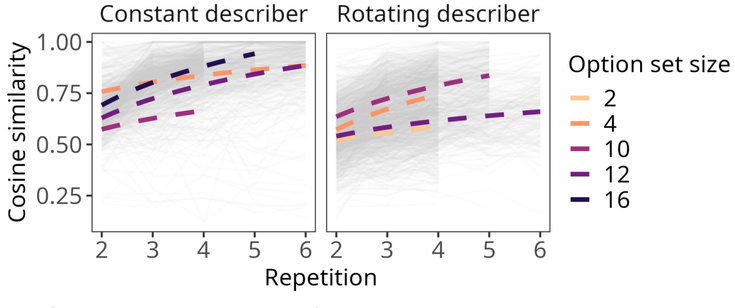
A goal of Refbank is to support systematic investigation of how experimental design factors moderate canonical reference game phenomena.



Reduction: Utterance length reduces over repetitions, but length also depends on group size and modality.



Parts of speech: The proportion of nouns increases across modality, but the proportion of closed class words only decreases in written exchanges.



Semantic trajectory: Similarity to previous description (measured with S-BERT embeddings) is higher if the describer is constant, but also varies by number of alternative image options.